

The smallest ellipse containing two unit disks has area $\frac{3\sqrt{3}}{2} \pi$

A rigorous proof, with a formally verified (Lean 4) companion

This note gives a complete, self-contained proof. The value $\frac{3\sqrt{3}}{2}\pi$ was found numerically by James Buddenhagen (2004) and recorded on Erich Friedman’s Packing Center, but does not appear to have been proved before. The argument is elementary — convexity of the ellipse, one monotonicity property of its quadratic form, one forced boundary point, and AM–GM; a final section records that it has also been formally verified in Lean 4, independently of which the mathematics stands on its own.

Theorem 1. *Among all ellipses containing two closed unit disks with disjoint interiors, the minimum area is*

$$A = \frac{3\sqrt{3}}{2} \pi \approx 8.1621,$$

attained by the axis-aligned ellipse with semi-axes

$$a = \frac{3\sqrt{2}}{2} = \frac{3}{\sqrt{2}} \text{ (major),} \quad b = \frac{\sqrt{6}}{2} = \sqrt{\frac{3}{2}} \text{ (minor),}$$

with the two disks centered at $(\pm 1, 0)$ on the major axis, touching each other at the origin. Each disk is internally tangent to the ellipse at two points, $(\pm \frac{3}{2}, \pm \frac{\sqrt{3}}{2})$.

(We say two disks are *interior-disjoint* if their interiors do not meet; for unit disks this is equivalent to their centers being at distance ≥ 2 .) The proof has four steps:

- **Step 1** (normalization): the ellipse may be taken axis-aligned and centered at the origin.
- **Step 2** (the main reduction): any ellipse holding two interior-disjoint unit disks also holds the single disk centered at $(1, 0)$ on the major axis.
- **Step 3** (lower bound): one boundary point of that disk, together with the AM–GM inequality, forces the area to be $\geq \frac{3\sqrt{3}}{2}\pi$, with equality only at $a^2 = \frac{9}{2}$, $b^2 = \frac{3}{2}$.
- **Step 4** (upper bound): that ellipse does contain two interior-disjoint unit disks, by a one-line perfect-square check.

Step 1. Normalization

Let E be any ellipse containing two interior-disjoint unit disks D_1, D_2 with centers p_1, p_2 . An ellipse is an affine image of a disk; in particular it has a center O_E (its center of symmetry) and two perpendicular axes of symmetry, the major and minor axes.

Apply the rigid motion (a rotation about O_E followed by a translation) that sends O_E to the origin and the major axis to the x -axis. Rigid motions preserve areas, the radii of the two disks,

and their disjointness, and carry E to an axis-aligned, origin-centered ellipse

$$E : \quad \frac{x^2}{a^2} + \frac{y^2}{b^2} \leq 1, \quad a \geq b > 0, \quad (\star)$$

where a is the semi-major and b the semi-minor axis, and $\text{Area}(E) = \pi ab$. (If E is a circle, $a = b$; the argument below specializes harmlessly, and we shall see it is not optimal.) Thus *without loss of generality every competitor has the form* $(\star)$, and we must

minimize πab subject to: E contains two interior-disjoint unit disks.

Throughout, write

$$u := a^2, \quad v := b^2, \quad u \geq v > 0.$$

Step 2. Reduction to the symmetric touching configuration

We prove that for a fixed ellipse $(\star)$, two interior-disjoint unit disks fit inside E if and only if the single unit disk centered at $(1,0)$ fits. The forward direction is the substance of this step; the converse is immediate. Consequently “ E holds two disjoint unit disks” reduces to the one-disk condition analyzed in Step 3, and the extremal configuration is the symmetric pair touching at the origin with centers on the major axis.

2a. The set of admissible centers is convex and doubly symmetric

Let

$$C := \{ z \in \mathbb{R}^2 : z + B(0,1) \subseteq E \}$$

be the set of centers of unit disks contained in E (the inner parallel body of E).

C is convex. Let $z_1, z_2 \in C$ and $0 \leq \lambda \leq 1$. For every $|w| \leq 1$ we have $z_1 + w \in E$ and $z_2 + w \in E$, so by convexity of E ,

$$(\lambda z_1 + (1 - \lambda) z_2) + w = \lambda(z_1 + w) + (1 - \lambda)(z_2 + w) \in E.$$

As this holds for all such w , $\lambda z_1 + (1 - \lambda) z_2 \in C$.

C is symmetric about both coordinate axes. The reflections $(x, y) \mapsto (-x, y)$ and $(x, y) \mapsto (x, -y)$ map E to itself, and carry the unit disk at z to the unit disk at the reflected center; so they preserve C . (C is nonempty iff $b \geq 1$, i.e. $v \geq 1$, the obvious requirement that one unit disk fit.)

2b. Feasibility forces a unit vector into C

Two unit disks are interior-disjoint iff their centers are at distance ≥ 2 . So suppose $p, q \in C$ with $|p - q| \geq 2$. By the triangle inequality $|p| + |q| \geq |p - q| \geq 2$, so at least one center — say p — satisfies $|p| \geq 1$. Since $O \in C$ (because $b \geq 1$) and $p \in C$, convexity gives the segment $[O, p] \subset C$; as $|p| \geq 1$, its point at distance 1 from O ,

$$m' := \frac{p}{|p|} \quad (\text{a unit vector, } |m'| = 1),$$

lies in C . Thus: *if E holds two disjoint unit disks, then C contains a unit vector m' , i.e. some unit disk with center on the unit circle fits in E .*

2c. The major-axis center is the easiest unit vector to contain

Intuitively, E is widest along its major (x -)axis, so a unit disk has the most room to fit when its center is pushed out along that axis; the major-axis center $(1, 0)$ should be the easiest unit vector to contain. We make this precise. For a center P , write $D_P = \{P + w : |w| \leq 1\}$ for the unit disk centered there. Since $E = \{Q \leq 1\}$ with $Q(x, y) = x^2/u + y^2/v$,

$$D_P \subseteq E \iff N(P) := \max_{Y \in D_P} Q(Y) \leq 1.$$

Lemma 1. *If $u \geq v$, then $N((1, 0)) \leq N(m')$ for every unit vector m' . Consequently, if the unit disk centered at some unit vector m' lies in E , so does the unit disk centered at $(1, 0)$.*

The proof rests on one elementary property of Q .

Radial monotonicity. In polar form $X = r(\cos \psi, \sin \psi)$,

$$Q(X) = r^2 \left(\frac{\cos^2 \psi}{u} + \frac{\sin^2 \psi}{v} \right) = r^2 \left(\frac{1}{v} - \left(\frac{1}{v} - \frac{1}{u} \right) \cos^2 \psi \right). \quad (1)$$

Since $u \geq v$, the coefficient $\frac{1}{v} - \frac{1}{u} \geq 0$, so at a fixed distance r from O , $Q(X)$ decreases as $\cos^2 \psi$ grows: Q is smallest along the x -axis and increases as X turns toward the y -axis. Thus two points equidistant from O satisfy $Q(X') \geq Q(X)$ whenever $\cos^2 \psi' \leq \cos^2 \psi$ — i.e. whenever X' makes the larger angle with the x -axis.

Proof of Lemma 1. Reflecting in the coordinate axes maps E to itself and a fitting disk to a fitting disk, and leaves N unchanged; so we may assume $m' = (\cos \varphi, \sin \varphi)$ with $\varphi \in [0, \frac{\pi}{2}]$. Let R be the rotation by φ about O . The unit ball is rotation-invariant, so R carries the disk at $(1, 0)$ onto the disk at $m' = R(1, 0)$:

$$D_{m'} = R D_{(1,0)}.$$

Let $B_0 \in D_{(1,0)}$ attain $N((1, 0)) = \max_{Y \in D_{(1,0)}} Q(Y)$, and write $B_0 = (1 + c, s)$ with $c^2 + s^2 \leq 1$. Replacing the offset (c, s) by $(|c|, |s|)$ gives the point $(1 + |c|, |s|)$, which is still in $D_{(1,0)}$ (the offset has the same length) and has Q at least as large: $(1 + |c|)^2 \geq (1 + c)^2$ and $|s|^2 = s^2$, so each term of Q is $\geq$ the original. Hence a maximizer with $c, s \geq 0$ exists, and we take $B_0 = (1 + c, s)$ with $c, s \geq 0$. Its polar angle ψ_0 then satisfies

$$\tan \psi_0 = \frac{s}{1 + c} \leq 1 \quad (\text{since } s \leq 1 \leq 1 + c),$$

so $\psi_0 \in [0, \frac{\pi}{4}]$.

Now $R B_0 \in R D_{(1,0)} = D_{m'}$, so $N(m') \geq Q(R B_0)$. The point $R B_0$ is at the same distance from O as B_0 , with polar angle $\psi_0 + \varphi$. Since $\psi_0 \leq \frac{\pi}{4}$ and $\varphi \leq \frac{\pi}{2}$,

$$\psi_0 \leq \psi_0 + \varphi \leq \psi_0 + \frac{\pi}{2} \leq \pi - \psi_0,$$

so $\psi_0 + \varphi$ lies in the band $[\psi_0, \pi - \psi_0]$, on which $\cos^2$ never exceeds $\cos^2 \psi_0$. By radial monotonicity (1), $Q(R B_0) \geq Q(B_0)$. Therefore

$$N(m') \geq Q(R B_0) \geq Q(B_0) = N((1, 0)),$$

and in particular $N(m') \leq 1$ forces $N((1, 0)) \leq 1$. □

2d. Conclusion of the reduction

Combining 2b and 2c: if E holds two disjoint unit disks, then C contains a unit vector m' , and by the Lemma the disk at $(1, 0)$ then lies in E . Conversely, if the disk at $(1, 0)$ lies in E then by the $x \mapsto -x$ symmetry so does the disk at $(-1, 0)$, and these two unit disks are interior-disjoint (centers distance 2 apart). Therefore:

For the fixed ellipse $(\star)$, E holds two interior-disjoint unit disks $\iff$ the unit disk centered at $(1, 0)$ lies in E .

The right-hand condition no longer mentions the disk positions. Thus, for bounding the area, we may replace “ E holds two disjoint unit disks” by the single condition that the disk centered at $(1, 0)$ lies in E ; Steps 3 and 4 analyze exactly this condition.

Step 3. The lower bound, from one boundary point and AM–GM

Let E be any ellipse containing two interior-disjoint unit disks. By Step 1 we may assume the normalized form $(\star)$, $E = \{x^2/u + y^2/v \leq 1\}$ with $u = a^2 \geq v = b^2$, and by Step 2 the unit disk centered at $(1, 0)$ lies in E . We need only *one* point of that disk. The boundary point

$$\left(\frac{3}{2}, \frac{\sqrt{3}}{2}\right) = (1, 0) + \left(\frac{1}{2}, \frac{\sqrt{3}}{2}\right), \quad \text{at distance 1 from } (1, 0),$$

lies in E , so

$$\frac{(3/2)^2}{u} + \frac{(\sqrt{3}/2)^2}{v} \leq 1, \quad \text{i.e.} \quad \frac{9}{u} + \frac{3}{v} \leq 4.$$

Set $X = \frac{9}{u}$ and $Y = \frac{3}{v}$, so $X, Y > 0$ and $X + Y \leq 4$. By the AM–GM inequality,

$$XY \leq \left(\frac{X+Y}{2}\right)^2 \leq 4.$$

But $XY = \frac{27}{uv}$, so $\frac{27}{uv} \leq 4$, i.e. $uv \geq \frac{27}{4}$, and therefore

$$\text{Area}(E) = \pi ab = \pi \sqrt{uv} \geq \pi \sqrt{\frac{27}{4}} = \frac{3\sqrt{3}}{2} \pi.$$

Equality forces $X = Y$ and $X + Y = 4$, i.e. $X = Y = 2$: then $\frac{9}{u} = 2$ and $\frac{3}{v} = 2$, giving $u = \frac{9}{2}$ and $v = \frac{3}{2}$. This is the only ellipse that can attain the bound.

Step 4. The construction attains it

It remains to exhibit a valid configuration of area $\frac{3\sqrt{3}}{2}\pi$. Take $u = a^2 = \frac{9}{2}$, $v = b^2 = \frac{3}{2}$ and the disks centered at $(\pm 1, 0)$ (so $ab = \sqrt{27/4} = \frac{3\sqrt{3}}{2}$, and the two disks are at distance 2, with disjoint interiors). By the $x \mapsto -x$ symmetry of E it suffices to check the disk at $(1, 0)$: a boundary point is $(1 + c, s)$ with $c = \cos t$, $s = \sin t$, and there

$$Q = \frac{(1+c)^2}{u} + \frac{1-c^2}{v} = \frac{(1+c)^2}{9/2} + \frac{1-c^2}{3/2} = -\frac{4}{9}c^2 + \frac{4}{9}c + \frac{8}{9} =: g(c).$$

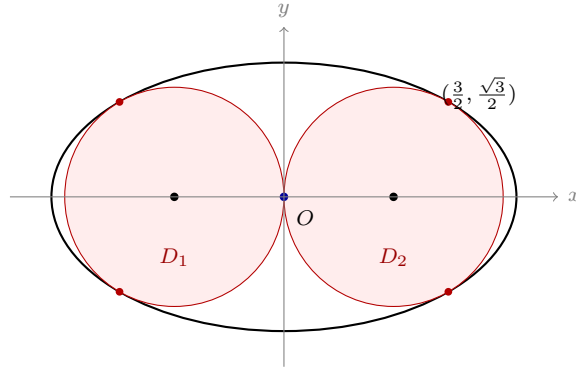
Multiplying $1 - g(c)$ by $uv = \frac{27}{4} > 0$ gives the perfect square

$$\frac{27}{4}(1 - g(c)) = 3c^2 - 3c + \frac{3}{4} = 3\left(c - \frac{1}{2}\right)^2 \geq 0,$$

so $g(c) \leq 1$ for all $c \in [-1, 1]$: the disk at $(1, 0)$, and by symmetry the disk at $(-1, 0)$, lies in E . Equality holds exactly at $c = \frac{1}{2}$ (i.e. $t = \pm\frac{\pi}{3}$), the four tangency points $(\pm\frac{3}{2}, \pm\frac{\sqrt{3}}{2})$; the two disks touch at the origin. Thus this ellipse genuinely contains two interior-disjoint unit disks and has area $\frac{3\sqrt{3}}{2}\pi$.

Combining Steps 3 and 4: the minimum area is exactly $\frac{3\sqrt{3}}{2}\pi$, and the minimizing ellipse is $\frac{x^2}{9/2} + \frac{y^2}{3/2} \leq 1$, unique up to rigid motion. ■

Figures



The optimal ellipse $x^2/\frac{9}{2} + y^2/\frac{3}{2} = 1$ with the two unit disks centered at $(\pm 1, 0)$, touching at O . Each disk is internally tangent to the ellipse at two points $(\pm\frac{3}{2}, \pm\frac{\sqrt{3}}{2})$ (red dots).

Remarks

Uniqueness of the ellipse. The equality analysis in Step 3 pins the optimal ellipse exactly: AM–GM equality and the boundary constraint together force $u = \frac{9}{2}$, $v = \frac{3}{2}$. So the minimizing ellipse $\frac{x^2}{9/2} + \frac{y^2}{3/2} \leq 1$ is unique up to rigid motion.

Where the difficulty lies. Once Step 2 is in hand, the bound is short: the single forced point $(\frac{3}{2}, \frac{\sqrt{3}}{2})$ and one application of AM–GM give it, with no optimization. The only genuinely geometric content is Step 2 — that for an ellipse with horizontal major axis, if two disjoint unit disks fit at all, then the disk centered at $(1, 0)$ fits. Naive global estimates do not suffice for the constant: the two disks have combined area $2\pi \approx 6.28$, well below $\frac{3\sqrt{3}}{2}\pi \approx 8.16$, so the bound cannot come from an area comparison alone.

Formalization in Lean 4

The argument above is self-contained and stands on its own. It has, in addition, been formalized and machine-checked in Lean 4 (with mathlib); the kernel reports the proof depends on no `sorry` and only the three standard axioms `[propext, Classical.choice, Quot.sound]`. We record the correspondence for the interested reader; nothing below is needed to follow the proof.

The Lean development treats a *fully general* ellipse $(x - c)^\top M(x - c) \leq 1$ with $M = \begin{pmatrix} A & B \\ B & C \end{pmatrix}$ positive definite (arbitrary tilt $B \neq 0$ and arbitrary center c); Step 1’s normalization is itself a kernel-checked rigid-motion reduction rather than an assumption. The capstone theorem is

$$\text{area_toReal_ge_of_feasibleGen} : \quad \frac{3\sqrt{3}}{2} \pi \leq (\text{volume}(\text{ellipseGenSet } A \ B \ C \ c)).\text{toReal},$$

where the area is the genuine Lebesgue measure and feasibility is literal interior-disjoint containment of two unit disks. A companion theorem exhibits the optimum ($\text{feasible_optimal} : \text{Feasible } \frac{9}{2} \ \frac{3}{2}$), so the bound is attained and the value is pinned exactly.

Step in this proof	Corresponding Lean content
Step 1 (normalization to $(\star)$)	rigid-motion reduction in General.lean : exists_diagonalizing (rotate away B via an eigenvector) and the rigid map rigidT , feeding area_ge_of_feasibleGen_matrix (area and disjointness preserved)
Step 2 (one ellipse holds two disjoint unit disks $\iff$ it holds the disk at $(1, 0)$)	Reduction.lean : convexity/symmetry of the admissible-center set and the major-axis comparison (Lean uses a cone-restricted rotation identity in place of the $\psi_0 \leq \pi/4$ argument; same conclusion)
Step 3 (one boundary point + AM–GM $\Rightarrow uv \geq \frac{27}{4}$)	uv_ge_of_fits (the Lean route reaches $uv \geq \frac{27}{4}$ by an equivalent sum-of-squares step rather than AM–GM)
Step 4 (perfect-square construction at $u = \frac{9}{2}, v = \frac{3}{2}$)	feasible_optimal , with tangency certificate $\frac{27}{4}(1 - g) = 3(c - \frac{1}{2})^2$
Area = $\pi\sqrt{uv}$	volume_ellipseSet via addHaar_image_linearMap

The two divergences noted in the table (the rotation identity in Step 2 and the sum-of-squares in Step 3) are independent routes to the same intermediate facts; both the hand proof and the formalization establish exactly the theorem stated above. The two algebraic identities used here — the boundary-point inequality and the tangency certificate — were additionally checked by exact symbolic computation.